YEAR 9 PHYSICS · UNIT BOOKLET

Work Like a Physicist

Collecting, processing and presenting data — eight lessons on how to produce evidence that other people can trust.

"The first principle is that you must not fool yourself — and you are the easiest person to fool."

Richard Feynman, physicist

NAME _____ CLASS _____

TEACHER _____ LAB PARTNER _____

KEEP THIS BOOKLET FOR ALL EIGHT LESSONS · BRING IT TO EVERY PRACTICAL

START HERE

What this booklet is for

Read this page once. It explains the whole unit in about two minutes.

Anyone can have an opinion about how the world works. A physicist has to do something harder: produce evidence that convinces people who were not there and who may not want to believe them.

That is the real skill in this unit — not equations, evidence. Over eight lessons you will learn to take a measurement you can defend, spot the errors hiding inside your own results, and turn a page of messy numbers into a graph that makes the pattern obvious.

HOW EVERY LESSON WORKS

Each lesson follows the same four-part shape, so you always know where you are:

The Case — a real story where this skill mattered. · Your Mission — what you should be able to do by the end. · The Tasks — numbered, in order. · Checkpoint — questions to prove you got it.

The eight lessons

Lesson 1	**Can this data be trusted?**	Uncertainty, precision, accuracy, resolution, errors
Lesson 2	**What are you actually changing?**	Variables, data types, choosing the right graph
Lesson 3	**The drop test**	Shock absorber practical · bar charts
Lesson 4	**Higher ramp, further flight?**	Ramp practical · line graphs
Lesson 5	**The point that does not fit**	Best-fit lines, anomalous results, graph quality
Lesson 6	**Solo flight**	Paper helicopter · independent challenge
Lessons 7–8	**Your investigation**	Design, run, graph and defend your own experiment
Reference	**The Physicist's Toolkit**	Every rule, formula and checklist in one place — page 3

ONE HABIT WORTH BUILDING NOW

When something in your results looks wrong, the useful question is never “can I ignore that?” It is “why did that happen?” Almost everything interesting in this unit is hiding inside that question.

There is a progress tracker on the last page of this booklet. Fill in a row at the end of every lesson.

REFERENCE

The Physicist's Toolkit

Everything you need for all eight lessons. Fold the corner of this page — you will come back to it.

The three variables

Independent	The one thing you change
Dependent	The thing you measure
Control	Everything you keep the same

Choosing your graph

Look at the independent variable only:

Continuous	Numbers on a scale — time, length, mass → **line graph**
Categoric	Names or groups — material, colour → **bar chart**

The two calculations

$$\text{mean} = \text{total of trials} \div \text{number of trials}$$

$$\text{uncertainty} = (\text{largest} - \text{smallest}) \div 2$$

Words that earn marks

Instead of "it went up", write "*as the ramp height increased, the distance travelled increased*". Instead of "it was wrong", name the error: *random* or *systematic*. Instead of "weird result", write "*possible anomalous result at 20 cm*".

Precision vs accuracy

Precise	Repeats are close *to each other*
Accurate	Close *to the true value*

You can be perfectly precise and completely wrong. That is the whole problem.

The two errors

Random	Scatters repeats unpredictably. **Repeats help.**
Systematic	Shifts every reading the same way. **Repeats do not help.**

Graph quality checklist

Title says what against what	☐
Independent variable on the x-axis	☐
Both axes labelled with units	☐
Sensible scale that fills the space	☐
Small crosses, plotted accurately	☐
Best-fit line, not dot-to-dot	☐
Bars equal width, with gaps	☐
Any odd point circled and explained	☐

THE RULE THAT DECIDES YOUR GRAPH

Continuous independent variable → line graph
Categoric independent variable → bar chart

It depends on what you *changed*, not on what you measured.

LESSON 1

Can this data be trusted?

Uncertainty · precision · accuracy · resolution · random and systematic error

THE CASE

The loose cable that broke physics

In September 2011, a team of physicists in Italy announced something that should have been impossible. They had timed neutrinos travelling 730 km from Switzerland, and the particles seemed to arrive about 60 nanoseconds *earlier* than light would. Nothing is meant to beat light.

The team had repeated the measurement thousands of times and the results were beautifully consistent. Newspapers around the world reported that Einstein had been proved wrong.

A few months later the team traced the problem to their own equipment — including a fibre-optic cable that was not properly connected. Every reading had been shifted by the same tiny amount, in the same direction, so repeating the measurement could never reveal it. The result was withdrawn.

Consistent does not mean correct. Today you learn the difference.

YOUR MISSION

- Explain why every measurement carries some uncertainty
- Calculate uncertainty from repeated readings
- Tell the difference between precision and accuracy — and prove it with an example
- Say what the resolution of an instrument is
- Describe random and systematic errors, and explain why repeats only fix one of them

1 Which data would you trust?

5 MIN

Three students timed the same toy car rolling down the same ramp. All three took three readings.

Data set	Trial 1 / s	Trial 2 / s	Trial 3 / s	Mean / s
A	1.20	1.23	1.19	1.21
B	1.05	1.47	1.12	1.21
C	0.90	0.93	0.89	0.91

a Sets A and B have *exactly* the same mean. Which one would you rather rely on, and why?

b Set C is the tidiest of the three. Explain how it could still be the most wrong.

c What is the point of taking three readings instead of one?

2 Precision and accuracy are not the same word

KEY IDEA

Precision = how close your repeats are to each other.
Accuracy = how close you are to the truth.
Four combinations are possible. All four happen in real labs.

Complete the last column. For each row, invent a one-line example from a school lab.

Situation	What the results look like	Example from a school lab
Accurate and precise	Tight cluster, on the true value	
Precise but not accurate	Tight cluster, in the wrong place	
Accurate on average, not precise	Widely spread, but the mean is close	
Neither	Widely spread and in the wrong place	

a Which of the four describes the neutrino result in *The Case*? Justify your choice.

3 How big is the doubt?

WORKED EXAMPLE

Every measurement has some doubt attached to it. With repeated readings you can put a number on that doubt.

FOLLOW EVERY STEP

Three readings: 1.20 s, 1.25 s, 1.21 s

Step 1 — Mean. (1.20 + 1.25 + 1.21) ÷ 3 = 3.66 ÷ 3 = 1.22 s

Step 2 — Range. maximum and minimum values = from 1.20 s to 1.25 s

Step 3 — Uncertainty. (largest − smallest) ÷ 2 = 0.05 ÷ 2 = 0.025 s, rounded sensibly to 0.03 s

Step 4 — Write it properly. mean time = 1.22 s ± 0.03 s

Read that last line out loud as: *"somewhere around 1.22 seconds, give or take 0.03 — and I am being honest about it."*

4 Your turn

10 MIN

Complete the table, then show one full mean calculation underneath.

Set	Trial 1	Trial 2	Trial 3	Mean	Range	Uncertainty
A	12.0	12.4	12.1			
B	8.4	9.1	8.6			
C	24.9	25.1	25.1			
D	3.20	3.65	3.25			

a Show your working for one mean:

b Most precise set: _____ Largest uncertainty: _____

c Which set would you insist on repeating before using it in a report, and why?

5 The two errors

KEY IDEA

RANDOM ERROR

Pushes readings up and down unpredictably. Your reaction time on a stopwatch is the classic one — sometimes early, sometimes late.

Repeats and a mean reduce it.

SYSTEMATIC ERROR

Shifts every reading the same way. A ruler measured from its end instead of the zero mark. A balance nobody zeroed. A loose cable.

Repeats do nothing. You have to find it.

a Explain, in your own words, why taking more repeats cannot rescue you from a systematic error.

Label each problem R (random) or S (systematic). One is deliberately arguable — be ready to defend it.

R/S	Problem	R/S	Problem
	Stopwatch started slightly late each time		Ruler starts measuring from its very end
	Judging bounce height by eye		Balance reads 0.2 g with nothing on it
	A draught from an open window		Reading a scale at an angle, always the same side

6 Resolution: what the instrument can even see

KEY IDEA

Resolution is the smallest change an instrument can detect. A stopwatch reading to 0.01 s cannot tell you about a 0.001 s difference — that information simply does not exist in your data.

Instrument	Resolution	Could it detect a change of 0.5 mm?
Ruler marked every 1 cm		
Ruler marked every 1 mm		
Stopwatch reading to 0.01 s		not a length
Balance reading to 0.1 g		not a length

TRAP

Better resolution gives you more decimal places. It does *not* make you right. A digital balance with a systematic fault will give you a wonderfully precise wrong answer to three decimal places.

7 Mini practical — measure something three times

15 MIN

Your teacher will give you one of these: a dropped-ruler reaction test, a toy car on a ramp, the bounce height of a ball, or one object measured with different rulers.

What I measured (with unit)	Trial 1	Trial 2	Trial 3	Mean	Uncertainty

a Instrument used, and its resolution:

b Name one random error that affected you, and say how you know it was random:

c Name one possible systematic error. How would you go looking for it?

8 Data detective

STRETCH

Three claims. For each, decide what you would need to know before believing it, and name the error you are most suspicious of.

The claim	What I would need to know first	Suspected error
"Our reaction times improved after training." The two tests were timed by different people.		
"This spring stretches 4.0 cm per 100 g." One reading was taken for each mass.		
"The classroom is exactly 6.000 m long." Measured once, with a tape held slightly slack.		

CHECKPOINT — LESSON 1

Use these words: **precise · accurate · random · systematic · the largest and smallest readings**

1. A set of results that are close together is _____.

2. A result that is close to the true value is _____.

3. Uncertainty is calculated using _____.

4. Repeats reduce the effect of _____ errors but do not correct _____ errors, because

_____

5. One thing I would do differently to make my data more repeatable:

_____

_____

TAKE IT HOME

Find a claim with a number in it — an advert, a headline, a sports statistic. Write down one question you would have to ask before you believed it, and bring it to Lesson 2.

_____

_____

LESSON 2

What are you actually changing?

Variables · continuous and categoric data · choosing the right graph

THE CASE

The study that proved nothing

Imagine an energy drink company runs a trial. Students who drank their product scored higher in a maths test than students who did not. The advert writes itself: drink this, get smarter.

Except the drinkers took the test in the morning and the non-drinkers took it after lunch. They were different students, in different rooms. One group had been told they were being studied.

The drink was changed — but so was everything else. When more than one thing changes at a time, no result can tell you which change caused what. The study is not weak evidence. It is no evidence.

Change one thing. Measure one thing. Keep everything else the same.

YOUR MISSION

- Identify the independent, dependent and control variables in any investigation
- Decide whether data is continuous or categoric
- Choose a line graph or a bar chart — and justify it using the right word

1 Take the question apart

10 MIN

Fill in all three columns. The control variable column is the one people rush — do not rush it.

Investigation question	I change (independent)	I measure (dependent)	I keep the same (two controls)
How does ramp height affect trolley speed?			
Which material is the best thermal insulator?			
How does the number of elastic bands affect launch distance?			
Which drink contains the most sugar?			

a Pick one of the four. If you failed to control the variables you listed, what exactly would go wrong with your conclusion?

2 Two kinds of data

KEY IDEA

CONTINUOUS

Numbers that can take any value on a scale, including the values in between. Length, time, temperature, mass.

Between 5 and 10 there is a 7.3 — and it means something.

CATEGORIC

Names, types or groups. Material, colour, brand, surface.

There is nothing halfway between “sponge” and “cardboard”.

LOOK ONLY AT WHAT YOU CHANGED

Continuous independent variable → line graph
Categoric independent variable → bar chart

THE MISTAKE ALMOST EVERYONE MAKES

Your dependent variable is nearly always a number. That is not the deciding factor. “Which surface gives the most friction?” measures distance in centimetres — but the graph is still a bar chart, because surface type is what you changed, and surfaces are categories.

3 Sort the investigations

10 MIN

Write LINE or BAR — then, in the last column, name the independent variable and label it continuous or categoric. Both parts are needed for full credit.

Investigation	Graph	Because the independent variable is…
How does temperature affect dissolving time?		
Which surface gives the most friction?		
How does mass affect the stretch of a spring?		
Which paper towel absorbs the most water?		
How does pendulum length affect its time period?		
Which material is the best sound absorber?		

4 Demonstration — which surface slows the car most?

CLASS DEMO

Your teacher rolls the same car down the same ramp onto different surfaces: bench, carpet, rubber mat, sandpaper.

Independent variable _____ Dependent variable _____ Graph type _____

a List three control variables — and star the one that would be genuinely hard to keep the same:

5 Fix the broken investigation

STRETCH

A student writes: "I will test whether expensive tennis balls bounce better. I'll drop an expensive ball from the window and a cheap one from my hand, and see which bounces higher."

a Name two things wrong with this plan:

b Rewrite it as a proper investigation question in the form "How does X affect Y?"

c What graph would your rewritten version need, and why?

CHECKPOINT — LESSON 2

1. The independent variable is the thing I _____.
2. The dependent variable is the thing I _____.
3. Use a line graph when _____.
4. Use a bar chart when _____.
5. Explain to someone who missed the lesson why graph choice depends on the independent variable and not the dependent one:

TAKE IT HOME

Write five investigation questions of your own — at least two needing a bar chart and two needing a line graph. For each, name the independent variable and label it continuous or categoric.

LESSON 3

The drop test

Shock absorber practical · repeats and means · bar charts

THE CASE

Why your helmet is made of polystyrene

A cycle helmet is mostly cheap foam. It is not there to be tough — it is there to be *crushed*. By squashing over a few thousandths of a second, the foam stretches out the time your head takes to stop, and that lowers the force on your skull.

Nobody takes a manufacturer's word for this. Before a helmet can be sold it is strapped to an instrumented head-shaped weight and dropped onto a steel anvil. Sensors record the impact. The test is repeated on helmet after helmet, because one lucky drop proves nothing.

Today you run the same kind of test, with a ball and a set of materials, and you make the same kind of decision: which material takes the hit best?

Same drop. Same ball. Same height. Only the material changes.

YOUR MISSION

- Recognise a categoric independent variable and explain what follows from that
- Collect three trials for each material and calculate a mean
- Build a results table with correct headings and units
- Draw a bar chart that would survive a strict marker
- Name a random and a systematic error in your own practical

1 Before you touch any equipment

5 MIN

Investigation question: How does the type of material affect the bounce height of a ball?

Independent variable	
Dependent variable (with unit)	
Three control variables	
Continuous or categoric?	
Graph type, and why	

THINK BEFORE YOU ANSWER

A good shock absorber soaks up energy instead of returning it. So does the best material give the *highest* bounce or the *lowest*? Decide now, and write it here: _____

2 Method

PRACTICAL

Work in your group. One person drops, one measures, one records — then swap so everyone does each job.

1	Choose 4 or 5 materials to test. Write them into the results table below.
2	Place the first material flat on the bench or in a tray.
3	Hold the ball at the same drop height every time — agree on it now: _____ cm
4	Release the ball. Do not throw it or push it down.
5	Measure the height of the first bounce, eye level with the top of the bounce.
6	Repeat three times for that material, then move on to the next.

SAFETY

Do not throw the ball. Drop vertically only. Keep the drop zone clear and stand back — do not crowd. Make sure the tray and materials are stable before you release.

3 Results

20 MIN

Write your material names in the first column. Record every reading as you take it — never from memory at the end.

Material	Bounce 1 / cm	Bounce 2 / cm	Bounce 3 / cm	Mean bounce / cm

a Show one mean calculation in full:

b Did any single reading make you stop and look twice? Which one, and what did you do about it?

4 Draw your bar chart

20 MIN

Material on the x-axis, mean bounce height on the y-axis. Equal-width bars, with gaps between them — the gaps are there to show that the categories are separate things, not points on a scale.

Title: _____

y-axis: mean bounce height / cm · x-axis: material

Check your own bar chart before your teacher sees it	✓
Title says what against what	☐
Material type on the x-axis	☐
y-axis labelled with the unit	☐
Scale goes up in sensible steps and uses most of the space	☐
Bars are all the same width, with equal gaps	☐
Bar heights match the mean values, not trial 1	☐

5 What the data says

10 MIN

a Which material was the best shock absorber? Quote a number from your results to back it up.

b Why did we repeat each drop three times instead of once?

Now pin down the errors. Be specific about *this* practical — "human error" earns nothing.

Error type	What went wrong in our version	How we would reduce it
Random		
Systematic		

c Would repeating the experiment ten more times have fixed the systematic error you named? Explain.

6 Push it further

STRETCH

a Another group tested the *same* materials and got completely different numbers. Give two reasons why that could happen without either group having done anything wrong.

b Suppose you tested foam of thickness 1 cm, 2 cm, 3 cm and 4 cm instead of different materials. What graph would you need now, and what changed?

CHECKPOINT — LESSON 3

1. I used a bar chart because the independent variable was _____.
2. The best shock absorber was _____ because it gave the _____ bounce.
3. One possible random error was _____.
4. One thing I would change if I ran this again:

LESSON 4

Higher ramp, further flight?

Ramp practical · continuous data · plotting a line graph properly

THE CASE

Nobody guesses a stunt jump

When a stunt team sends a car off a ramp, the driver does not find out how fast to go by trying it. They build the ramp, run the car at several different speeds, measure how far it lands each time, and plot the results.

The graph is the point. Once you have a line through five measured points, you can read off the speed needed for a distance you have *never actually tested*. That is what a line graph buys you: a prediction, from a pattern.

A bar chart cannot do that. A single measurement certainly cannot. One dot on a page is not a pattern — it is an anecdote.

Five points and a line will tell you about the gaps in between.

YOUR MISSION

- Recognise a continuous independent variable and explain what follows from it
- Build a results table with five values, three trials each, and a mean
- Choose a scale that uses the whole grid instead of a corner of it
- Plot points accurately as small crosses
- Label both axes with the quantity *and* the unit

1 Set it up

5 MIN

Investigation question: How does the height of a ramp affect the distance a trolley travels after it leaves the ramp?

Independent variable	
Dependent variable	
Continuous or categoric?	
Graph type	

Four control variables. Tick each one off as your group agrees on it.

SAFETY

Check the ramp is stable before every release. Never aim a trolley at a person. Keep bags and stools well away from the run. Collect the trolley by hand — do not chase it.

2 Method

PRACTICAL

1	Set the ramp to your first height. Use five heights — 5, 10, 15, 20 and 25 cm work well.
2	Place the trolley at the same marked starting point on the ramp every time.
3	Release it. Do not push — a push adds energy you cannot measure or repeat.
4	Measure the distance travelled from the bottom of the ramp to where the trolley stops.
5	Repeat three times at that height, then change the height and repeat everything.

WORK FAST, BUT WRITE EVERYTHING DOWN

You have limited time. The single biggest time-waster is having to redo a height because nobody wrote down trial 2. Record each number the moment you read it.

3 Results

25 MIN

Ramp height / cm	Distance 1 / cm	Distance 2 / cm	Distance 3 / cm	Mean distance / cm
5				
10				
15				
20				
25				

a Largest mean: _____ cm Smallest mean: _____ cm

b You need those two numbers before you can choose a scale. Explain why.

4 Choose your scale first

DO NOT SKIP

More marks are lost to bad scales than to bad measuring. Work this out *before* you plot a single point.

HOW TO CHOOSE A SCALE

1. Find your largest value.
2. Count the big squares available on that axis.
3. Divide, then round *up* to something easy: 1, 2, 5, 10, 20, 50 per square.
4. Check your largest value still fits.

NEVER DO THIS

Scales going up in 3s, 7s or 15s — nobody can read them.

Squeezing all your data into the bottom-left quarter of the grid.

Changing the size of the steps halfway up an axis.

a My y-axis will go from 0 to _____ cm, with each big square worth _____ cm.

b My x-axis will go from 0 to _____ cm, with each big square worth _____ cm.

5 Plot your line graph

20 MIN

Ramp height on the x-axis, mean distance on the y-axis. Mark each point with a small ×, not a blob — a fat dot hides where the point really is. Leave the line until Lesson 5.

Title: _____ Remember the units on both axes

6 Mistake hunt

10 MIN

These are the six things markers look for first. Check your own graph honestly, then swap with a partner and check theirs.

Common mistake	Mine	Partner's
Units missing from one or both axes	☐	☐
Awkward scale nobody can read	☐	☐
Graph crammed into one corner of the grid	☐	☐
Dependent variable put on the x-axis by mistake	☐	☐
Bars drawn instead of plotted points	☐	☐
Points joined dot-to-dot	☐	☐

a One thing my partner's graph does better than mine:

CHECKPOINT — LESSON 4

1. The independent variable was _____, measured in _____.

2. The dependent variable was _____, measured in _____.

3. A line graph was the right choice because _____.

4. The control variable that was hardest to keep the same was _____, because

5. Looking at your points so far — do you expect the pattern to be a straight line or a curve? Commit to an answer, and say why.

LESSON 5

The point that does not fit

Best-fit lines · anomalous results · graph quality · writing a conclusion

THE CASE

The readings that were too strange to believe

In 1985 three scientists from the British Antarctic Survey published something alarming: every spring, the ozone layer above Antarctica was collapsing. They had spotted it from readings taken on the ground, year after year.

Satellites had been watching the same sky. The values they recorded were so far below anything expected that the processing software flagged them as readings it could not trust. The scientists did not bin them, and did not believe them straight away either: they checked them against measurements taken from the ground.

The ground-based results were published first. A few months later the satellite team showed their maps, and the hole was there in the record.

An odd result is a question, not rubbish. Ask it before you bin it.

YOUR MISSION

- Draw a best-fit line — straight or curved — that shows the trend
- Explain why dot-to-dot is wrong for experimental data
- Identify a possible anomalous result and say what you would do about it
- Improve a graph against a checklist
- Write a conclusion supported by measurements

1 Which graph is better?

5 MIN

Your teacher will show you three versions of the same data: dot-to-dot, a best-fit straight line, and a best-fit curve.

a Which one shows the pattern most clearly, and what makes it better?

b What is a dot-to-dot line actually claiming about the measurements? Why is that claim untrue?

2 How a best-fit line works

KEY IDEA

A best-fit line shows the trend, not every measurement.

It should have roughly as many points above it as below it. It does not have to touch a single one of them.

DO

Draw one smooth line — straight or gently curved — in pencil, with a ruler if it is straight.

Aim for a balance of points either side.

Extend it across the range of your data.

DO NOT

Zig-zag from point to point.

Force the line through the first and last points and ignore the middle.

Draw a thick, wobbly, double-stroke line — precision matters here too.

a Your results are "rarely perfect". Explain what a best-fit line is admitting about experimental data.

3 Spot the anomalous result

10 MIN

Another class ran the ramp experiment and got these means.

Ramp height / cm	Mean distance / cm
5	20
10	37
15	52
20	96
25	80

Plot these five points on the grid, then draw a best-fit line through the trend.

a Which value looks wrong, and what exactly makes it look wrong?

b Should the class delete it? Write what they should do *first*, and why.

c Give two things that could have caused a reading like this in the ramp practical:

THE HONEST ANSWER

An anomalous result that turns out to be a measuring mistake gets corrected or repeated. An anomalous result you cannot explain gets reported, not hidden — circled on the graph with a note. Deleting data because it is inconvenient is the one thing a scientist must never do.

4 Finish your own graph

20 MIN

Go back to your ramp graph from Lesson 4. Add the best-fit line. Circle anything that does not sit near it and write a short note beside it. If there are no anomalous results, write "no anomalous results — all points lie close to the line". Saying so is worth marks.

Before you call it finished	✓
Title included	☐
x-axis labelled with unit	☐
y-axis labelled with unit	☐
Sensible scale, using most of the grid	☐
Points plotted accurately as small crosses	☐
Best-fit line drawn — not dot-to-dot	☐
Anomalous results circled and commented on, or their absence stated	☐

5 Peer review

10 MIN

Swap booklets. Mark your partner's graph against the list above, then write them one specific improvement — not "make it neater".

Reviewed by	
Strongest thing about this graph	
One specific improvement	

6 Write the conclusion

15 MIN

Use these sentence starters. Where it says "the graph shows", quote a real number from your results — that is the difference between a conclusion and an opinion.

> As the ramp height increased, the distance travelled by the trolley _____

> I know this because the graph shows _____
>
> _____

> For example, at _____ cm the mean distance was _____ cm, while at _____ cm it was _____ cm.

> One possible source of random error was _____

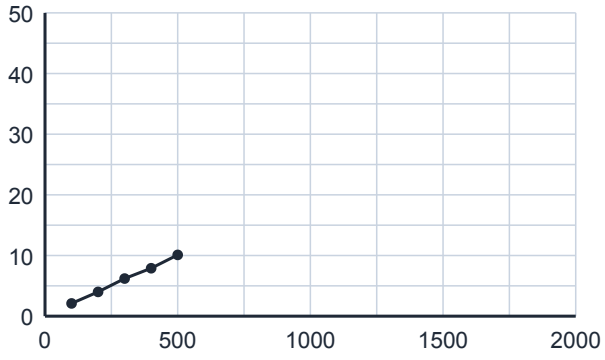
> One result that may be an anomalous result is _____ because
>
> _____

7 Mark someone else's graph

STRETCH

A student hands in the graph below for the spring experiment. It has four separate problems. Find them, mark them on the graph, and list them.

Mass / g	Extension / cm
100	2.1
200	4.0
300	6.2
400	7.9
500	10.1



a The four problems:

_____

_____

_____

b Which one costs the most marks, and why?

_____

c Redraw the student's graph properly here, with a best-fit line.

CHECKPOINT — LESSON 5

1. A best-fit line is _____.
2. We avoid dot-to-dot lines because _____.
3. An anomalous result is _____.
4. Before ignoring an anomalous result you should _____.
5. Why would deleting an inconvenient result be dishonest rather than just untidy?

TAKE IT HOME

Find a graph somewhere outside school — a news site, a textbook, a sports app, a product advert. Run it through the graph quality checklist on page 3.

a Where the graph came from, and what it shows:

b Two things it does well:

c One thing that would lose marks if a Year 9 student handed it in:

LESSON 6

Solo flight

Paper helicopter challenge · the whole process, on your own

THE CASE

The helicopter that flew on Mars

In April 2021 a small helicopter called Ingenuity lifted off the surface of Mars. Getting it to fly was brutally hard: the Martian atmosphere is around one per cent as dense as Earth's, so the rotors have almost nothing to push against.

The engineers could not just try it and see. They tested rotor designs in a chamber pumped down to Martian pressure, changing blade size and spin rate, measuring lift, plotting the results, and reading the pattern off the graph. Ingenuity's rotors ended up over a metre across for a craft with a mass under 2 kg.

Today you do a much smaller version of the same job — with paper. But the process is identical, and this time nobody is going to walk you through it.

This one is on you. Booklet, ruler, brain.

YOUR MISSION — INDEPENDENTLY, FROM START TO FINISH

- Identify all three types of variable
- Design your own results table with units
- Collect three trials for each of five values and calculate means
- Choose the correct graph and justify it
- Plot accurately, add a best-fit line, deal with anomalous results
- Write a conclusion supported by your own numbers

1 Plan before you cut anything

10 MIN

Recommended question: How does wing length affect the fall time of a paper helicopter? If your teacher lets you choose, you could instead investigate paperclip mass or drop height.

My research question	
Independent variable (and unit)	
Dependent variable (and unit)	
Three control variables	
Is the independent variable continuous or categoric?	
Which graph will I draw, and why?	
Five values I will test	

SAFETY

Do not climb on chairs, stools or benches to gain drop height. Use a height you can reach standing on the floor and use the same one every time. Take care with scissors. Keep the landing zone clear.

2 Design your own results table

ASSESSED

Draw it yourself in the space below. Nothing is printed for you here on purpose. A good table has: a column for the independent variable, three trial columns, a mean column, and units in every heading.

3 Fly and record

25 MIN

Build each helicopter, add the same paperclips every time, and drop from the same height. Time the fall three times before changing anything.

GETTING DECENT TIMINGS

The same person should time every drop — swapping timers introduces a new error partway through.

Start the watch as the helicopter is released, not as you let go of the watch.

Fall times are short, so your reaction time matters a lot. That is exactly why you repeat.

IF A DROP GOES WRONG

If it hits a desk, catches a draught, or somebody knocks your arm — say so, and take that trial again. Note it down. A discarded trial with a written reason is honest. A quietly deleted one is not.

a Any trials you repeated, and why:

4 Graph your results

ASSESSED

Everything you have learned so far, in one graph. Title, axes, units, scale, accurate points, best-fit line, anomalous results noted.

Title:

Check both axes have units before you hand this in

5 Conclusion

ASSESSED

As _____ increased, the fall time _____

The evidence for this is _____

_____

A specific pair of numbers from my results: at _____ the mean was _____, at _____ it was _____.

My graph _____ contain a possible anomalous result. Details: _____

a Try a prediction. Using your best-fit line only, what fall time would you expect for a wing length you did *not* test? State the value and where you read it from.

6 Mark yourself

SELF-ASSESSMENT

Be strict. This is the same list your teacher will use.

Success criteria	Not yet	Done
Independent, dependent and control variables all identified	☐	☐
Table designed by me, with units in every heading	☐	☐
Three trials for every value of the independent variable	☐	☐
Means calculated correctly	☐	☐
Correct graph type, with a reason given	☐	☐
Axes labelled, units shown, sensible scale	☐	☐
Best-fit line rather than dot-to-dot	☐	☐
Anomalous results identified, or their absence stated	☐	☐
Conclusion quotes evidence from my own results	☐	☐

CHECKPOINT — LESSON 6

1. The most important thing I have learned about drawing graphs is

2. The most important thing I have learned about collecting data is

3. One thing I would do differently next time, and the reason:

TAKE IT HOME — EVALUATION PRACTICE

Write four sentences about your investigation: one strength of your method, one weakness, one possible random error, and one improvement you would actually make. You will need all four in the project that starts next lesson.

LESSONS 7–8

Your investigation

Design it · run it · graph it · defend it

THE CASE

The scientists who tried to prove themselves wrong

In 2015 the LIGO detectors picked up a signal from two black holes colliding a billion light years away. It was the first direct detection of gravitational waves, and it won a Nobel Prize.

The team did not announce it. For months they attacked their own result — checking instruments, ruling out interference, running the analysis again. They had good reason to be careful: the collaboration had previously used *blind injections*, secret fake signals slipped into the data to test whether the team would spot them and whether they would follow their own procedures properly.

Only once they had failed to break their own result did they publish it.

Your job over these two lessons: produce a result you cannot break.

YOUR MISSION

- Write a testable question in the form “How does X affect Y?”
- Plan a safe, fair investigation and get it approved
- Collect repeated measurements and calculate means
- Choose and draw the right graph for your own data
- Write a conclusion backed by evidence and an honest evaluation

1 The rules

READ FIRST

EVERY GROUP MUST

Work in a pair or a three.
Ask one question: “How does X affect Y?”
Change one independent variable only.
Measure one dependent variable.
Identify at least three control variables.

AND EVERY GROUP MUST

Test at least four or five values or categories.
Take at least three trials of each.
Calculate a mean for each.
Choose the correct graph type.
Write a conclusion *and* an evaluation.

2 Choose a question that will work

10 MIN

Some starting points — you may use one of these or invent your own.

Research question	Independent variable	Likely graph
How does ramp height affect the distance travelled by a trolley?	Ramp height	Line
How does wing length affect the fall time of a paper helicopter?	Wing length	Line
How does surface type affect the distance travelled by a toy car?	Surface type	Bar
How does material type affect the bounce height of a ball?	Material type	Bar
How does the number of elastic bands affect launch distance?	Number of bands	Line

And some that will waste your two lessons. Work out what is wrong with each one before your teacher tells you.

Rejected question	What is wrong with it
“What is the best paper helicopter?”	
“How does height affect it?”	
“How does surface affect speed and distance and time?”	
“How does throwing force affect distance?”	

3 Planning grid

LESSON 7

Our research question	
Group members	
Independent variable (and unit)	
Dependent variable (and unit)	
Continuous or categoric?	
Graph we will use, and why	
At least three control variables	
Values or categories we will test (4–5)	
Equipment we need	
Safety risks	
How we will reduce those risks	

4 Write your method

LESSON 7

Numbered steps. Someone from another class should be able to follow it without asking you a single question. Include how many trials you will take and how you will keep the control variables the same.

5 Teacher checkpoint

BEFORE ANY EQUIPMENT

STOP HERE

Do not collect any data until your teacher has signed this box. This is not a formality — in a real lab, no experiment runs without approval.

Check	✓	Check	✓
Question is clear and testable	☐	Method is safe	☐
Independent variable is clear	☐	Equipment is available	☐
Dependent variable is measurable	☐	Repeats are included	☐
Control variables are realistic	☐	Graph choice is sensible	☐

Approved by		Date	

6 Results table

LESSON 7

Design your own headings. If your independent variable is categoric, replace the first heading with the category name — "Surface type", "Material", and so on.

a Show one mean calculation in full:

WHILE YOU COLLECT

Record results the moment you take them. Put units in the headings, not next to every number. Repeat each measurement at least three times. If a result looks odd, check it now — you will not get the equipment back.

6b Lab notes

Anything that happened during data collection that a reader would want to know: equipment that misbehaved, a trial you repeated, a control variable that slipped, a surprise. Write it here as it happens — you will not remember next lesson, and this is where your evaluation comes from.

CHECKPOINT — END OF LESSON 7

1. Our research question is _____.
2. Our independent variable is _____ and it is _____.
3. Our dependent variable is _____.
4. One control variable we genuinely managed to keep the same was _____.
5. One result we may need to check next lesson is _____.

7 Decide your graph — again

LESSON 8

Do not draw anything yet. Answer these five first.

Is your independent variable continuous or categoric?	
So: line graph or bar chart?	
What goes on the x-axis?	
What goes on the y-axis?	
What units are needed, and where?	

a Largest mean value: _____ So each big square on the y-axis will be worth _____

8 Draw your graph

LESSON 8

Title:

Crosses for a line graph · gaps between bars for a bar chart

9 Interrogate your own results

LESSON 8

a What pattern does the graph show? Does the dependent variable increase, decrease, or stay about the same?

b Is the pattern strong and clear, or weak and scattered? What in the graph tells you that?

c Any possible anomalous results? Say which, or state clearly that there are none.

d What is the single strongest piece of evidence in your results — the one number a doubter would find hardest to argue with?

10 Conclusion

ASSESSED

Use the version that matches your investigation.

> If your graph is a line graph:
> As _____ increased, _____ _____
> I know this because _____

> If your graph is a bar chart:
> The category with the highest _____ was _____
> I know this because _____

> Both: the numbers that prove it are _____
> _____

> This suggests that _____

11 Evaluation

ASSESSED

This is where the marks hide. Be specific — name the actual thing that went wrong, not "human error".

> One strength of our method was _____

> One weakness of our method was _____

> One possible random error was _____

> One possible systematic error was _____

> If we repeated this investigation we would improve it by _____
> _____

a How confident are you in your conclusion — and what would it take to make you more confident?

12 Peer review

LESSON 8

Swap with another group. Their job is to find the weakness you missed. Yours is to be grateful for it.

Feature	✓	Feature	✓
Question uses "How does X affect Y?"	☐	Correct graph type chosen	☐
Independent variable identified	☐	Axes labelled correctly	☐
Dependent variable identified	☐	Units shown on the graph	☐
Three or more control variables	☐	Sensible scale used	☐
Results table includes units	☐	Points or bars accurate	☐
At least three trials	☐	Best-fit line where appropriate	☐
Means calculated	☐	Anomalous results considered	☐
Conclusion uses evidence	☐	Evaluation names an improvement	☐

Reviewed by	
The strongest part of this investigation	
The question I would ask this group	

13 Going further

STRETCH

a Compare your results with another group who investigated something similar. Do you agree? If not, what could explain the difference?

b Is your relationship linear (a straight line) or non-linear (a curve)? What does that tell you?

c What range of values would you test if you had another two lessons, and why that range?

CHECKPOINT — END OF THE UNIT

1. The most important decision in our investigation was _____.

2. The strongest evidence from our results was _____.

3. The biggest source of error was probably _____.

4. What I personally contributed to the group:

5. The hardest part of the whole unit, and what I would tell next year's Year 9 about it:

TAKE IT HOME — INDIVIDUAL REFLECTION

Answer these on your own, not as a group. Your teacher wants your view, not the group's.

1 What did your group investigate, and what did your graph show?

2 What did you personally contribute?

3 What was the most difficult part, and how did you handle it?

4 If a stranger read your conclusion, what is the first objection they would raise?

GLOSSARY

Every word you need

If you use these words correctly, you sound like a physicist. If you avoid them, you lose marks.

Word	What it means
Accuracy	How close a measurement is to the true value
Anomalous result (anomaly)	A result that does not fit the pattern of the others
Bar chart	The graph you use when the independent variable is categoric
Best-fit line	A single straight or curved line showing the overall trend, with points balanced either side
Categoric data	Data sorted into named groups or types — material, colour, brand
Continuous data	Number data that can take any value on a scale — length, time, mass
Control variable	Something you deliberately keep the same so it cannot affect the result
Dependent variable	The thing you measure
Independent variable	The one thing you change
Line graph	The graph you use when the independent variable is continuous
Mean	The average: the total of the trials divided by the number of trials
Precision	How close repeated measurements are to each other
Random error	Unpredictable variation between repeats. Repeats and a mean reduce it
Range	The maximum and minimum values of the independent or dependent variables
Repeatable	The same person, method and equipment give similar results again
Resolution	The smallest change a measuring instrument can detect
Systematic error	An error that shifts every reading the same way. Repeats do not fix it
Trial	One repeat of a measurement
Uncertainty	The interval within which the true value can be expected to lie

Phrases that pick up marks

Instead of writing…	Write…
“It went up.”	“As the ramp height increased, the distance travelled increased.”
“It was human error.”	“Reaction time when starting the stopwatch caused random error.”
“One result was weird.”	“The value at 20 cm is a possible anomalous result — it does not fit the trend.”
“The results were good.”	“The repeats were close together, so the data is precise.”
“We did it three times.”	“Three trials were taken at each value so that a mean could be calculated.”
“It's a bar chart.”	“A bar chart was used because material type is categoric.”

Track your progress

Tick a row at the end of every lesson. Be honest — this is for you, not for a mark.

L	I can…	Not yet / Getting there / Solid
1	Work out uncertainty and tell precision from accuracy	○ ○ ○
2	Identify all three variables and choose the right graph	○ ○ ○
3	Take repeat readings, find a mean, draw a bar chart	○ ○ ○
4	Plot an accurate line graph with labelled axes and units	○ ○ ○
5	Draw a best-fit line and deal with an anomalous result properly	○ ○ ○
6	Run a whole investigation on my own	○ ○ ○
7 – 8	Design my own experiment and defend the conclusion	○ ○ ○

THE WHOLE UNIT IN ONE SENTENCE

Measure it more than once, be honest about the doubt, show the pattern clearly, and let the evidence decide.

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